



10. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$ is equal to

(1) 4

(2) 3

(3) 2

(4) $\frac{1}{2}$

Answer:

We know that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x \frac{\sin 4x}{\cos 4x}} \times \frac{(3 + \cos x)}{1}$$

$$= \lim_{x \rightarrow 0} \frac{2 \frac{\sin^2 x}{x^2}}{\frac{\sin 4x}{4x}} \times \frac{(3 + \cos x)}{4} = 2$$

Correct choice is (3) 2